# VIMS IRMS Facility

Virginia Institute of Marine Science

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## Rates & Analytical Services
## September 2027 – Cost per Sample

For inquiries regarding GC-FID, GC-IRMS, EA-IRMS, or GB-IRMS analyses, please contact
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# EA-IRMS

Elemental Analyzer optimized for stable carbon and nitrogen isotopic analyses in solid matrices.

***Instrument***: Delta-V advantage IRMS (Thermo Fisher Scientific) with dual inlet and Conflo IV interfaced with an elemental analyzer peripheral (EA; Costech ECS 4010 CHNS-O) for $\delta^{13}C$ and $\delta^{15}N$ of solid samples

| Analytical Service | Internal | External |
|---|---|---|
| C & N in solids | $9.25 | $10.92 |
| Tinning samples | $10.00 | $11.80 |
| Difficult samples | $3.00 | $3.54 |
| Acid fumigation | $2.25 | $2.66 |






# GB-IRMS

GasBench II continuous flow isotope ratio determinations of $CO_2$ and $N_2$ gases.

***Instrument***: Delta-V advantage IRMS (Thermo Fisher Scientific) with dual inlet and Conflo IV interfaced with a GasBench II peripheral (Thermo Fisher Scientific) for $\delta^{13}C$ and $\delta^{15}N$ of headspace gases and dissolved inorganic carbon (DIC).

| Analytical Service | Internal | External |
|---|---|---|
| $CO_2$ in gas | $17.00 | $20.06 |
| $N_2$ in gas | $12.50 | $14.75 |
| DIC | - | - |

# GC-IsoLink IRMS

Compound Specific Isotopic Analysis (CSIA) for lipid biomarkers and Amino Acids. If interested in other compounds, please contact us.

***Instrument***: A Trace 1610 GC and a GC IsoLink II (combustion/conversion) coupled via a ConFlo IV to a Delta Q IRMS (Thermo Fisher Scientific)

| Analytical Service | Internal | External |
|---|---|---|
| CSIA-FAMES & n-Alkanes | $140.50 | $165.79 |
| Biomarkers extraction | $71.50 | $84.37 |
| FAMES derivatization | $92.50 | $109.15 |
| GC-FID (quantification) | $16 | $18.88 |
| Amino acids | $250.00 | $295.00 |

